



Development of a representative driving cycle for urban buses based on the K-means cluster method

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Abstract

To establish a representative driving cycle for public urban buses in Fuzhou city, real-world driving data of 18 buses' routes was gathered. Based on acquisition of the real operational data, about 2.2 million seconds of valid data was divided into 8262 micro trips and an evaluation metric comprised of 15 characteristic parameters was established. Next, principal component analysis was applied for dimensionality reduction of the characteristic parameters and all micro trips were classified into different clusters by K-means clustering. Subsequently, the Silhouette equation was introduced as part of the cluster selection procedure. Finally, the target driving cycle, containing a 1227 second speed-time series, was developed according to a comprehensive estimation of six characteristic parameters and a maximum of the speed-acceleration frequency distribution. The experimental results verified the effectiveness and accuracy of the proposed method.

Keywords Driving cycle · Micro trip · Principal component analysis · K-means clustering · Silhouette equation

1 Introduction

A driving cycle is described by vehicle speed versus time to represent the real-world operation of a certain region or a certain vehicle type. Generally, two major categories of driving cycle are known as legislative and non-legislative, which has a wide range of applications on type approval of vehicles, improving the performance of electric vehicles [1,2].

Legislative driving cycles, such as WLTC, NEDC, FTP-75 and J10-15, are compulsorily executed by governments to certify a vehicle's emissions and fuel economy within their respective jurisdictions. Non-legislative driving cycles, such as the Hong Kong driving cycle [3], Seoul city bus driving cycle [4], Singapore driving cycle [5], are used in research for energy conservation and pollution evaluation.

Since driving patterns vary from city to city and from one area to another, the available driving cycles obtained

for certain cities or countries are not usually applicable for other cities and countries. What's more, the existing standard driving cycles cannot keep up with the changing road infrastructure, the changing vehicle fleet composition, and the growing number of vehicles on the road. Therefore, much research work is targeted towards developing new and representative driving cycles for their own city or region [6–9].

Fuzhou is the capital city of Fujian province, located on China's southeastern coast. Its urban population was about 3 million inhabitants as of the 2015 census. However, its vehicle population varied from 478 thousand of the year 2012 to 695 thousand of the year 2015 with the average annual growth 15%. As a typical medium scale city, there is no existing driving cycle in Fuzhou. The purpose of this investigation is to obtain a better understanding of the driving characteristics of urban buses in Fuzhou such as the idling proportion, average speed and acceleration.

The approach of developing the driving cycle for urban buses combines the methods of principal component analysis (PCA) and k-means clustering. The k-means clustering method is one of the most widely used techniques available in solving the clustering problem [7,10–12]. However, on selection of driving features and the number of clusters, most previous research relied on the comparison of different driving features such as idle proportions, average speed and

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acceleration artificially. It is not persuasive to describe how to determine the number of clusters in reasonable way.

The aim of the activity presented in this paper is to develop a representative driving cycle for urban buses in Fuzhou based on the K-means cluster method combined with principal component approach. Moreover, the Silhouette value was used to evaluate the appropriateness of clustering, and the error of artificial factors in clustering process was curbed.

2 Data collection

Route selection is a critical aspect in developing driving cycles. If routes selected can not reflect the typical traffic condition in the target area, the final driving cycle achieved will indicate a deviation from its actual situation. In urban area of Fuzhou city, there run 211 buses' routes. Finally 18 buses' routes (shown in Fig. 1) are selected on the following process:

- (1) For each bus station in Route R, accumulate the number of times to get the station frequency (F_s), which expresses how many bus lines stop at this bus station;
- (2) For each bus route, accumulate F_s of all its bus stations as the route's frequency (F_r).
- (3) The intensity of bus station for the bus route R is defined as $I_r = F_r/N_s$, herein N_s means the numbers of bus station for the bus route R.

Therefore, larger the value I_r is, stronger is the representativeness of the route.

The data collection device shown in Fig. 2, was connected with the vehicle's on-board diagnostic system (OBD) and

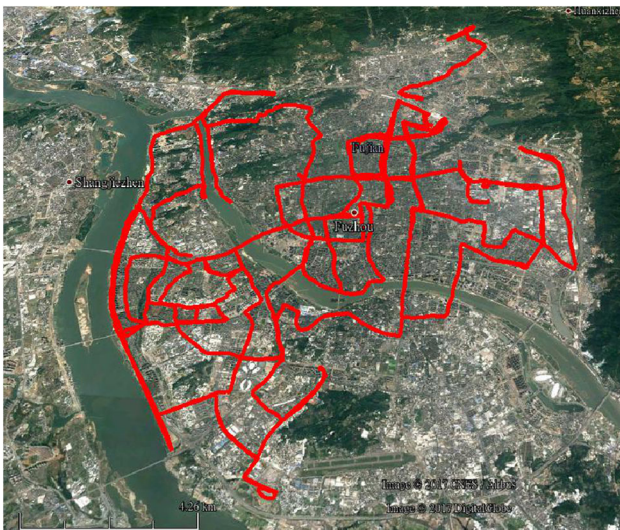


Fig. 1 Routes of the experimental vehicles



Fig. 2 Data collection device

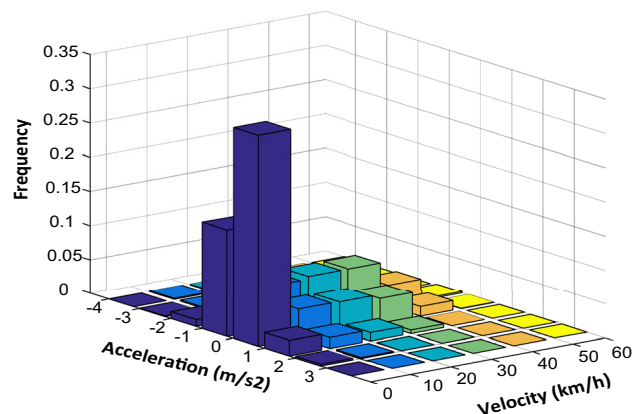


Fig. 3 The frequency of speed-acceleration distribution

used for data collection. During driving, this device can collect many parameters, including the speed of the vehicle, engine speed, torque percentage, and the instantaneous fuel consumption at a frequency of 1 HZ.

Before the construction of an initial driving cycle, the high acceleration points of 2.5 ms^{-2} or higher should be removed in order to reduce GPS error caused by vibration. In this study, after the removal of the acceleration data, there was about 2.2 million seconds of valid data for modeling. After data processing, the frequency of speed-acceleration distribution was determined, as shown in Fig. 3. Figure 4 shows the curve of speed vs time for the sample data, and the curve of acceleration vs time for the data is shown in Fig. 5.

Fig. 4 The curve of speed versus time for the sample data

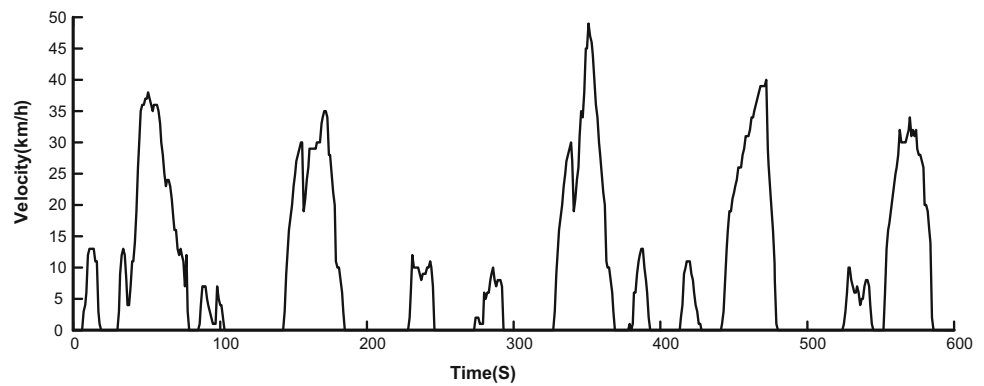


Fig. 5 The curve of acceleration versus time for sampling data

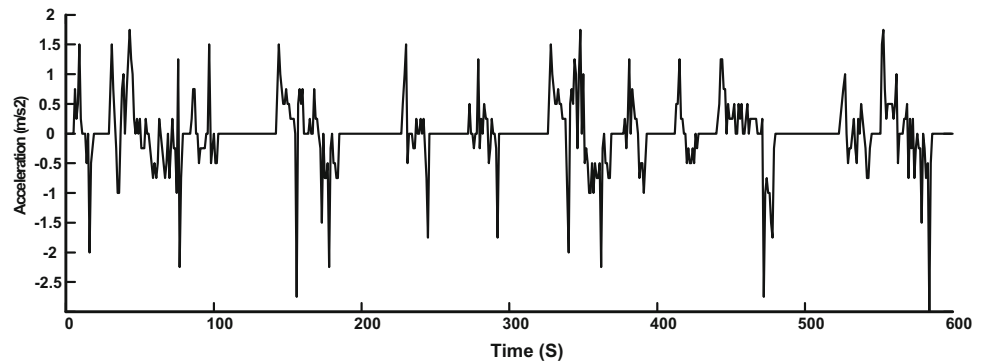
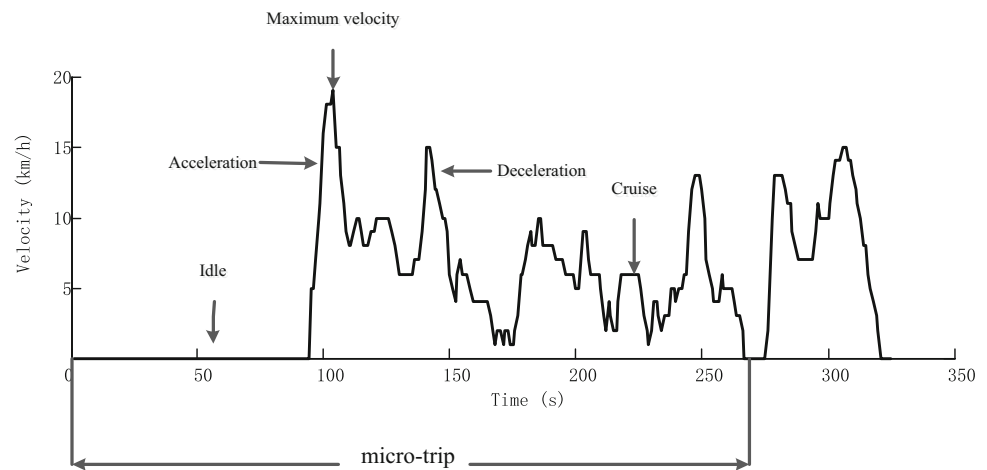


Fig. 6 Components of a micro trip



3 Micro trip segment

The vehicle driving state is strongly affected by traffic condition, and may include several starts and stops between the start point and the destination. The time between the start of a period of idle velocity to the next period of idle velocity is defined as a micro-trip. The data presented in Fig. 6 shows that an overall trip is composed of many smaller fragments, or micro-trips. Four driving components of a micro-trip are defined as follows:

- (1) Idling: a continuous process during which time the engine is working, but the vehicle speed is zero.
- (2) Acceleration: a continuous process when vehicle acceleration is greater than 0.1 m s^{-2} .
- (3) Cruising: a continuous process during which, the absolute vehicle acceleration is less than 0.1 m s^{-2} , but the vehicle speed is not zero.
- (4) Deceleration: a continuous process during which, the vehicle acceleration is less than 0.1 m s^{-2} .

Table 1 Characteristic parameters

Characteristic parameter	Definition	Unit
T	Segment duration	s
S	Travel distance	km
V _a	Mean velocity	km h ⁻¹
V _x	Mean run velocity	km h ⁻¹
a _{max}	Maximum acceleration	m s ⁻²
a _{min}	Maximum deceleration	m s ⁻²
T _i	Percent of time idling	%
T _a	Percent of time accelerating	%
T _d	Percent of time decelerating	%
T _C	Percent of time cruising	%
v _{max}	Maximum velocity	km h ⁻¹
V _{std}	Standard deviation of velocity	km h ⁻¹
a _a	Mean acceleration	m s ⁻²
a _{std}	Standard deviation of acceleration	m s ⁻²
a _d	Mean deceleration	m s ⁻²

Many characteristic parameters are used to describe each micro trip. In this study, each micro-trip is represented by a 15 element vector, as shown in Table 1.

Many characteristic parameters were acquired by MATLAB for each of the 8262 short travel fragments that were identified from the sample data. A subset of these is presented in Table 2.

4 Principal component analysis

PCA is a statistical procedure that uses an orthogonal transformation to convert a number of possibly correlated variables into a smaller number of uncorrelated variables called principal components, which keep as much of the vari-

ability in the original observations data as possible. Fifteen characteristic parameters in Table 1 attempt to establish the statistical metrics of driving cycle. However, some parameters may be strongly interrelated. The potential overlap of parameters increases the complexity of the required calculations, so PCA was used to reduce the size of state vectors for the statistical metrics. Standardization of the data is preferable when the variables are in different units. The formula is as follow:

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad (1)$$

where a_{ij} ($i = 1, 2, m; j = 1, 2, \dots, n$) is the micro trip number i for parameter number j .

Standardizing matrix X :

$$X_{mn} = \begin{bmatrix} \chi_{11} & \chi_{12} & \cdots & \chi_{1n} \\ \chi_{21} & \chi_{22} & \cdots & \chi_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \chi_{m1} & \chi_{m2} & \cdots & \chi_{mn} \end{bmatrix} \quad (2)$$

where

$$\chi_{ij} = a_{ij} - \bar{a}_j / s_j \quad (3)$$

$$\bar{a}_j = \frac{1}{m} \sum_{i=1}^m a_{ij} \quad (4)$$

$$s_j^2 = \frac{1}{m-1} \sum_{i=1}^m (a_{ij} - \bar{a}_j)^2 \quad (5)$$

The standardized cycle state vector can be used in PCA, with the formula as follows:

Table 2 Subset of characteristic parameters determined for each micro trip

No.	T/s	S/km	v _a /km h ⁻¹	...	T _i /s	T _a /s	T _C /s	V _{std}	a _{std}
1	54	0.118	7.833	...	37	11	2	13.0221	1.246
2	88	0.168	6.886	...	63	12	3	13.133	0.637
3	92	0.426	16.652	...	7	28	21	10.091	0.545
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
4130	136	1.117	29.559	...	20	32	46	16.015	0.507
4131	61	0.042	2.491	...	43	6	6	4.210	0.399
4132	131	0.402	11.053	...	76	18	20	15.195	0.532
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
8260	202	0.762	13.579	...	108	41	20	17.154	0.607
8261	232	0.7439	11.543	...	48	64	33	8.178	0.502
8262	52	0.032	2.211	...	34	6	7	3.913	0.306

Covariance matrix Σ :

$$\Sigma = \begin{bmatrix} s_1^2 & \text{cov}(1, 2) & \cdots & \text{cov}(1, n) \\ \text{cov}(2, 1) & s_2^2 & \cdots & \text{cov}(2, n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(n, 1) & \text{cov}(n, 2) & \cdots & s_n^2 \end{bmatrix} \quad (6)$$

where

$$s_x^2 = \text{cov}(x, x) \quad (7)$$

$$\text{cov}(x, y) = \text{cov}(y, x) = \frac{1}{m-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad (8)$$

Correlation matrix R:

$$R = \frac{1}{m-1} \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \quad (9)$$

where

$$r_{xy} = \frac{\text{cov}(x, y)}{s_x s_y} \quad (10)$$

λ_i represents the eigenvalues of matrix R, which $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$.

$\lambda_i / \sum_{j=1}^n \lambda_j$ is the contribution of principal component number i . The greater the contribution rate, the more information is presented.

Table 3 shows the contribution and accumulation rate of each principal component. It can be seen that eigenvalues of the first 4 principal components are greater than 1 and the accumulation rate is 86.797%. Theoretically, processing the principle component analysis could be done with a few principle components, who make an accumulation rate exceeding 80% [13]. To simplify the problem and reduce computation and save resources, the first 4 principal components were selected for subsequent analysis.

A larger absolute value of a parameter in the principal component load matrix indicates that this parameter has a closer correlation coefficient with the principal component. According to Table 4, the first principal component reflects maximum velocity, mean run velocity, the standard deviation of velocity, travel distance, mean velocity and the percent of time accelerating. The second principal component reflects the standard deviation of acceleration and segment duration. The third principal component reflects the percentage of time idling and the mean deceleration. The fourth principal component reflects mean acceleration.

Table 3 Contribution rate of principal components and the cumulative contribution rate

Component	Eigenvalues	Contribution rate (%)	Accumulation rate (%)
1	6.783	45.218	45.218
2	3.235	21.564	66.782
3	1.610	10.731	77.513
4	1.393	9.284	86.797
5	0.760	5.070	91.866
6	0.359	2.396	94.262
7	0.281	1.871	96.133
8	0.230	1.532	97.665
9	0.138	0.919	98.583
10	0.082	0.548	99.131
11	0.050	0.333	99.464
12	0.044	0.293	99.757
13	0.022	0.148	99.906
14	0.014	0.094	100.000
15	1.006E-013	1.040E-013	100.000

5 K-means clustering method

Cluster analysis is used to classify events of similar certain qualities into a category, and classify events with different qualities into other category. One of the most widely used clustering techniques available in the literature is the k-means algorithm.

The K-means clustering approach sets clusters K according to the understanding of clustering problem. A representative sample of each class is selected as an initial focal point, and the other samples are classified according to their distance from that focal point. In general, Squared Euclidian Distance was used to analyze the data [14], and has the formula as follows:

$$d_{ij} = \sum_{k=1}^p (x_{ik} - x_{jk})^2 \quad (11)$$

where, each micro trip has P parameters, x_{ik} is the micro trip of number i in parameter K , and d_{ij} is the distance of each micro trip.

Based on the four principal components contained from PCA, K-means clustering method was performed to cluster micro trips into two classes, three classes and four classes respectively, as shown in Table 5.

To obtain the best clustering result, the equation of silhouette was applied in this study. The silhouette value displays how close each point in a cluster is to points in neighboring clusters [15,16]. The equation is as follows:

Table 4 Principal component load matrix

Characteristic parameter	M1	M2	M3	M4
Segment duration	0.495	0.664	− 0.512	0.210
Travel distance	0.883	0.353	0.038	− 0.092
Mean velocity	0.879	− 0.167	0.321	− 0.093
Mean run velocity	0.919	− 0.158	0.002	0.117
Maximum acceleration	0.622	− 0.311	0.074	0.488
Maximum deceleration	− 0.467	0.512	0.490	0.380
Percent of time idling	− 0.074	0.403	− 0.760	0.468
Percent of time accelerating	0.810	0.422	− 0.044	− 0.263
Percent of time decelerating	0.769	0.481	0.206	0.042
Percent of time cruising	0.565	0.553	0.021	− 0.298
Maximum velocity	0.929	− 0.160	− 0.009	0.130
Standard deviation of velocity	0.896	− 0.227	0.021	0.106
Mean acceleration	0.183	− 0.603	0.061	0.615
Mean deceleration	− 0.079	0.557	0.602	0.341
Standard deviation of acceleration	0.507	− 0.793	− 0.090	− 0.184

Table 5 Results of K-means analysis

Micro-trip	Class 2	Class 3	Class 4
1	1	1	1
2	2	2	4
3	1	1	1
4	2	3	3
5	2	2	3
⋮	⋮	⋮	⋮
4129	1	2	2
4130	2	2	2
4131	2	3	3
4132	1	1	1
4133	1	1	4
⋮	⋮	⋮	⋮
8257	1	1	1
8258	1	2	4
8259	1	2	2
8260	2	3	4
8261	2	3	3
8262	1	1	1

$$L(i) = \frac{\min(\mathbf{b}) - a}{\max[a, \min(\mathbf{b})]} \quad (12)$$

where a is the average distance between the micro trip i with other micro trips in the same class. $\mathbf{b}$ is a vector, and is the average distance between the i micro trip with other micro trips in the different class.

$L(i)$ value ranges from -1 to 1 , indicating different cluster meaning for the micro trip i . When $L(i)$ is greater than 1 , the micro trip i is farther away from the other clusters. If $L(i)$ is equal to 0 , the micro trip i is not distinctly grouped into a specific cluster from other clusters. If $L(i)$ is less than 0 , the micro trip i is probably grouped into the wrong cluster.

The silhouette values of each micro trip in each class are shown in Fig. 8. As Figure 7a shows, the silhouette value of almost each micro trip was more than 0 when micro trips are clustered into two classes, indicating that most micro trips were well clustered. As Fig. 7b shows, the analysis showed that there were many negative silhouette values for micro trips as the number of clusters is 3 . As Fig. 7c shows, there were many negative silhouette values in first and fourth class when micro trips are clustered into 4 classes. In conclusion, class 3 and class 4 have many micro trips assigned to the wrong cluster. Therefore, the best number of clusters is selected as two on the basis of the silhouette values.

According to the result of clustering, statistics on the time proportion of four driving conditions, including idling, accelerating, cruising, decelerating, for the two clusters are shown in Fig. 8.

As can be seen from histogram, a high percentage of time is spent on idling, 40.48% and only around 13.79% time is occupied by normal driving operations for the first cluster, indicating that the vehicle drives in a congested traffic condition. In second cluster, there is less time idling, and the rest of time for other three driving operations are evenly distributed, indicating that the pattern corresponded to the vehicle driving in smooth traffic conditions.

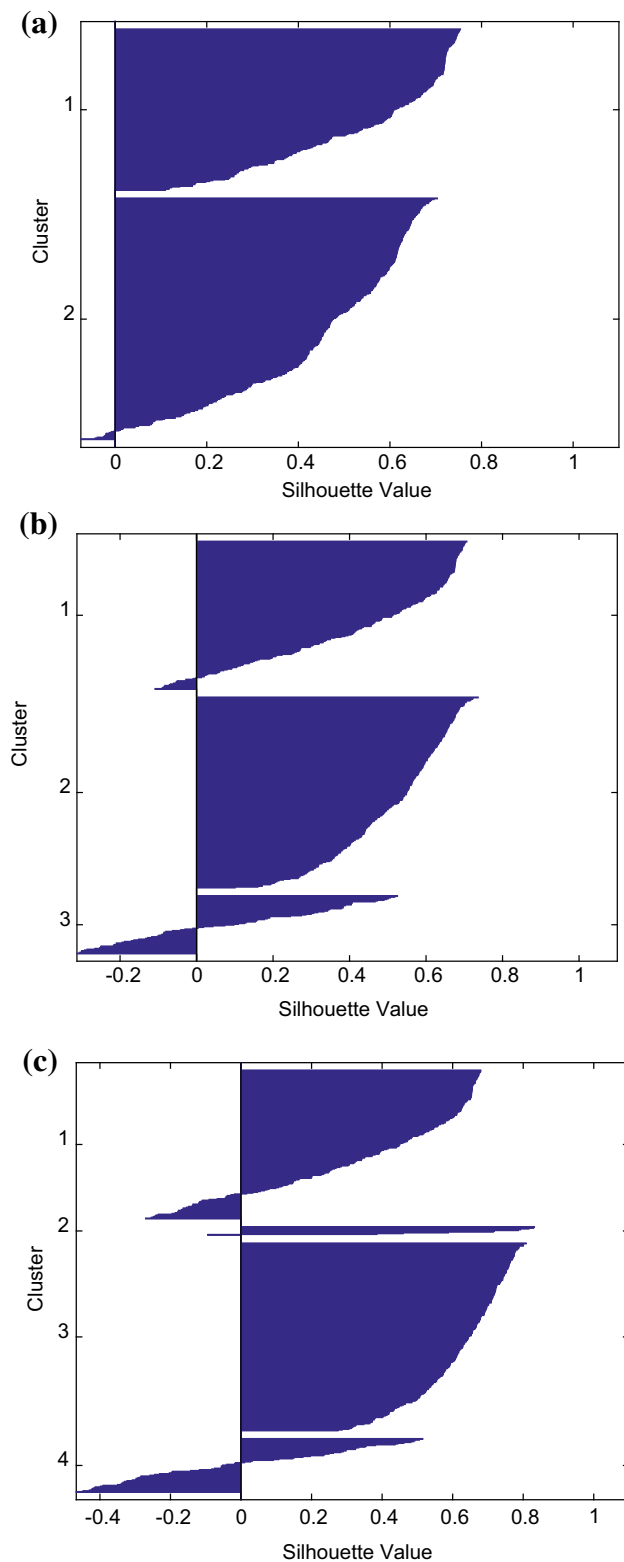


Fig. 7 The silhouette value for the different clusters classified. **a** 2 Clusters, **b** 3 clusters, **a** 4 clusters

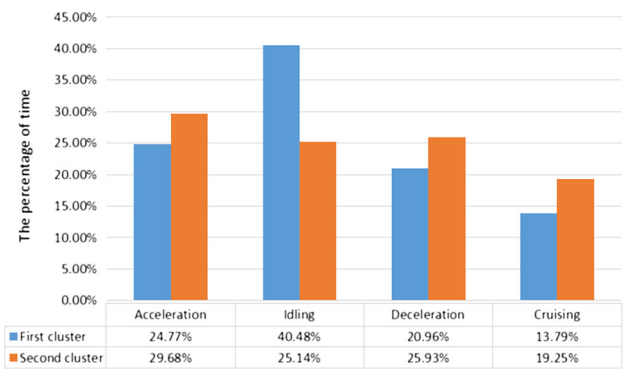


Fig. 8 Comparison on time proportion of four driving condition for two clusters

6 Driving cycle generation

K-means cluster analysis was used to determine the distance between each micro trip and the cluster center. As shown in equation (12). Smaller d_{ij} is, closer is the relationship of the micro trip and the class clustered. Therefore, the generation of candidate driving cycle was performed as follows. First, according to the size of cluster center in each type, 10 candidate micro trips were selected from smallest to largest [17,18]. If one type contained less than 10 micro trips, all micro trips of this type were selected as candidate micro trips. Next, based on different kinds of candidate micro trips, random selection was proposed to synthesize candidate driving cycles consisting of over 1200 second. Finally, a target driving cycle was achieved from candidate driving cycle through the following criteria.

There are different approaches to assess the target driving cycle. Some scholars use minimum performance value (PV) to choose the final driving cycle. In this study, 6 characteristic parameters (percentage of accelerating, percentage of idling, percentage of cruising, mean velocity, mean acceleration, and the standard deviation of acceleration) and the maximum difference of the speed-acceleration frequency distribution (SAFD) were applied on the selection of target driving cycle.

7 Results

We applied the method described above for the construction of the driving cycle, obtaining a target driving cycle 1227 second in length by MATLAB, as shown in Fig. 9, with the frequency of speed-acceleration distribution shown in Fig. 10.

The characteristic parameters of the target driving cycle and sample data, shown in Table 6, have relative errors within the range of $\pm 10\%$. The SAFD difference between the whole sample data and the target driving cycle is shown in Fig. 11, and reveals that the differences are within $\pm 4\%$. Thus, the

Fig. 9 The target driving cycle of 1240 s

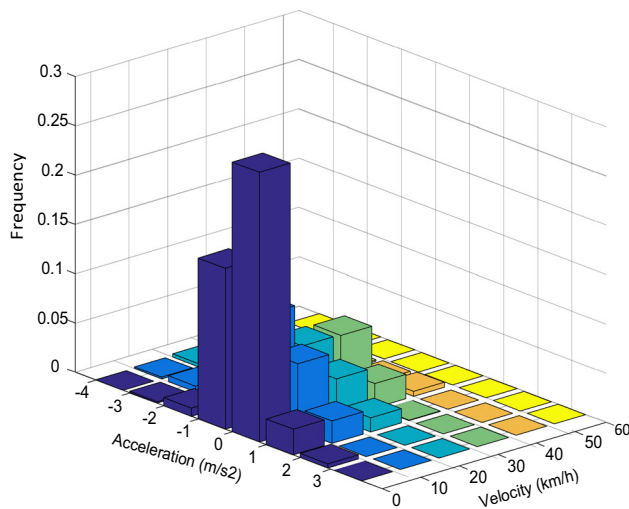
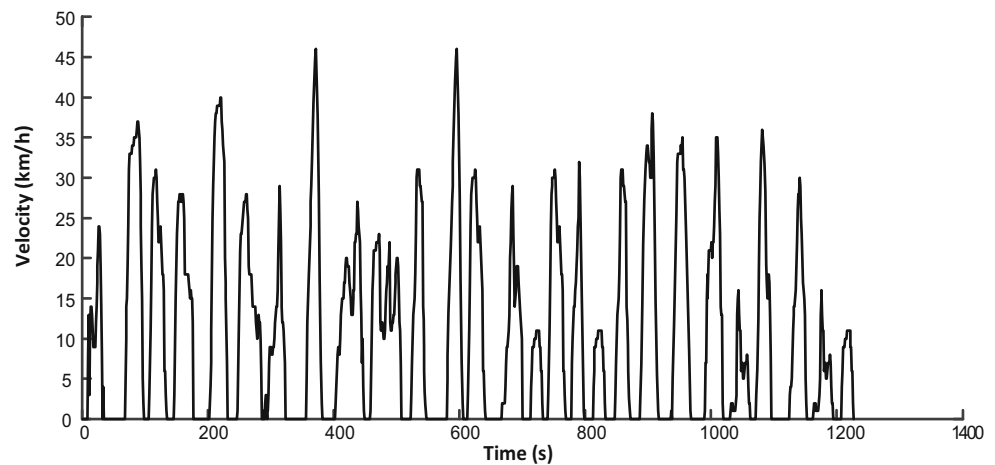


Fig. 10 The frequency of the speed-acceleration distribution

Table 6 Comparison of parameters for the whole sampling data and the target driving cycle

Characteristic parameter	Sample data	Target cycle	Variance (%)
T_a	0.255	0.270	6.04
T_c	0.149	0.155	3.77
T_i	0.376	0.344	8.64
V_a	12.353	13.816	11.85
a_a	0.713	0.740	3.76
a_{std}	0.492	0.493	0.21

target driving cycle well reflects the driving features of the sample data.

The comparison of Fuzhou urban buses driving cycle (FZBDC) and major standard driving cycles, shown in Table 7, reveals that the difference between FZBDC and other standard driving cycles is significant. The average speed of FZBDC is extremely slower than the cycle of UDSS, NEDC

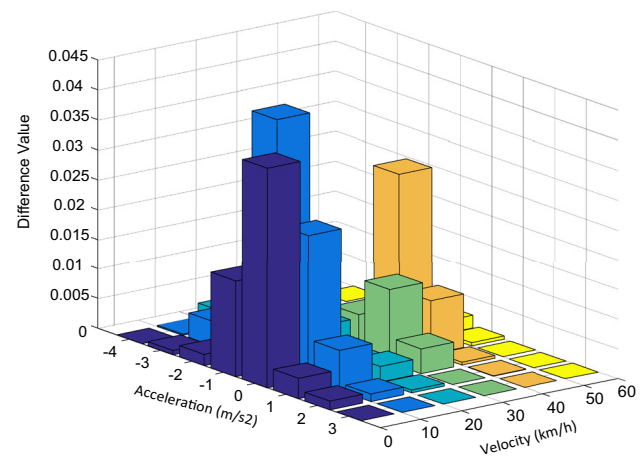


Fig. 11 SAFD difference between the overall sampling data and the target driving cycle

Table 7 Comparison of Fuzhou urban buses driving cycle with world-wide driving cycles

Driving cycles	T	T_a	T_i	T_c	V_a	a_a
FZBDC	1227	0.270	0.344	0.155	13.82	0.740
WHTC	1800	0.287	0.171	0.215	20.22	0.741
CBDC	1314	0.219	0.292	0.237	15.90	0.643
UDSS	1369	0.294	0.178	0.273	31.51	0.5
NEDC	1184	0.229	0.239	0.374	33.21	0.540
FTP-75	1875	0.393	0.181	0.077	34.10	0.169
J10-15	660	0.261	0.326	0.195	22.68	0.540

and FTP-75, but is closet to the Chinese Urban Buses Driving Cycle (CBDC). Besides that, the FZBDC has the longest time proportion of idling. The slowest average speed and longest idling time proportion may deliberately infer that the problems of fuel consumption, emissions and traffic congestion in Fuzhou is more serious with comparison of other city considering the rapid growing of vehicle population.

8 Conclusion

In this study, a new approach for constructing a representative driving cycle for urban buses is presented based on K-means cluster method combined with PCA. The silhouette value calculated was suggested to obtain reduce error during clustering and the best clustering results was obtained consequently. Candidate micro trips used for the construction of the candidate driving cycle were selected by the size of the cluster center. Finally, the target driving cycle was extracted by a comprehensive estimate of the 6 characteristic parameters and the maximum of the speed-acceleration frequency distribution. Experimental results demonstrated the effectiveness and the accuracy of the suggested method.

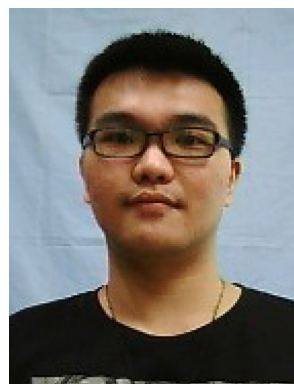
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References

- Berzi, L., Delogu, M., Pierini, M.: Development of driving cycles for electric vehicles in the context of the city of Florence. *Transp. Res. Part D* **47**, 299–322 (2016)
- Brady, J., O'Mahony, M.: Development of a driving cycle to evaluate the energy economy of electric vehicles in urban areas. *Appl. Energy* **177**, 165–178 (2016)
- Tong, H.Y., Hung, W.T., Cheung, C.S.: Development of a driving cycle for Hong Kong. *J. Atmos. Environ.* **33**, 2323–2335 (1999)
- Wi, H., Park, J., Lee, J., Kim, W. et al.: Development of a city bus driving cycle in Seoul based on the actual patterns of urban bus driving. SAE Technical Paper, 2009-01-2914. (2009). <https://doi.org/10.4271/2009-01-2914>
- Ho, S.H., Wong, Y.D., Chang, V.W.C.: Developing Singapore driving cycle for passenger cars to estimate fuel consumption and vehicular emissions. *Atmos. Environ.* **97**, 353–362 (2014)
- Kamble, S.H., Mathew, T.V., Sharma, G.K.: Development of real-world driving cycle: case study of Pune, India. *Transp. Res. Part D* **14**, 132–140 (2009)
- Fotouhi, A., Montazeri-Gh, M.: Tehran driving cycle development using the K-means clustering method. *Sci. Iran.* **20**(2), 286–293 (2013)
- Tamsanya, N., Chungpaibulpattana, S., Atthajariyakul, S.: Development of automobile Bangkok driving cycle for emissions and fuel consumption assessment. In: *Proceedings of the 2nd Joint International Conference on Sustainable Energy and Environment*, Bangkok, 21–23, November 2006 (2006)
- Amirjamshidi, G., Roorda, M.J.: Development of simulated driving cycles for light, medium and heavy duty trucks: case of the Toronto Waterfront area. *Transp. Res. Part D* **34**, 255–66 (2015)
- Datong, Q., Sen, Z., Zhenggang, Q., Shuijiang, C.: Driving cycle construction using K-means clustering method. *J. Jilin Univ. Eng. Technol.* **46**(2), 383–389 (2016)
- Zaccardi, J.M., Le Berr, F.: Analysis and choice of representative drive cycles for light duty vehicles-case study for electric vehicles. *Proc. Inst. Mech. Eng. Part D* **227**(4), 605–616 (2013)
- Prohaska, R., Konan, A., Kelly, K., Lammert, M.: Heavy-duty vehicle port drayage drive cycle characterization and development. In: *SAE Commercial Vehicle Engineering Congress (COMVEC)*. At Rosemont, Illinois (2016). <https://doi.org/10.4271/2016-01-8135>
- Dunteman, G.H.: *Principal Components Analysis*, pp. 22–23. Sage, New York. (1989) ISBN 0-8039-3104-2
- Singh, A., Yadav, A., Rana, A.: K-means with three different distance metrics. *Int. J. Comput. Appl.* (0975–8887) **67**(10), 13–17 (2013)
- Hamza, R., Muhammad, K., Arunkumar, N., González, G.R.: Hash based Encryption for Keyframes of Diagnostic Hysteroscopy. *IEEE Access* (2017). <https://doi.org/10.1109/ACCESS.2017.2762405>
- Fernandes, S.L., Gurupur, V.P., Sunder, N.R., Arunkumar, N., Kadry, S.: A novel nonintrusive decision support approach for heart rate measurement. *Pattern Recognit. Lett.* (2017). <https://doi.org/10.1016/j.patrec.2017.07.002>
- Arunkumar, N., Ramkumar, K., Venkatraman, V., Abdulhay, E., Fernandes, S.L., Kadry, S., Segal, S.: Classification of focal and non focal EEG using entropies. *Pattern Recognit. Lett.* **94**, 112–117 (2017)
- Arunkumar, N., Kumar, K.R., Venkataraman, V.: Automatic detection of epileptic seizures using new entropy measures. *J. Med. Imaging Health Inf.* **6**(3), 724–730 (2016)



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